Citation Equity: Testing GEO Writing Advice, and Keeping Only What Moved Citation

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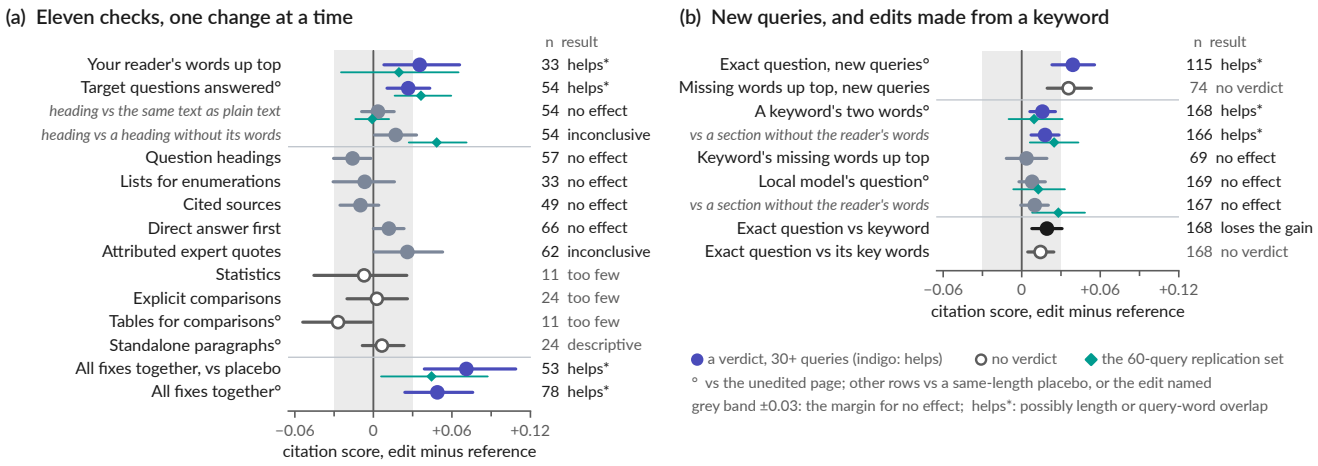


Figure 1: Of eleven GEO checks, only the reader's own words near the top moved citation; the gain held on new queries, and a keyword kept part of it. Difference in citation score from a same-length placebo or the unedited page (°), 95% intervals. (a) Each check alone on 80 GEO-bench queries (260 for checks with too few pages); two rows place the question section's gain in its words. (b) The same edits on 180 new queries, then made from a keyword or a local model's question; the bottom rows compare the exact question with the guesses.

ABSTRACT

Writers are told that question headings, lists, statistics, quotes and cited sources help AI answer engines cite their pages. We tested eleven such checks by editing real web pages one change at a time, each against a same-length paraphrase, while open models answered from fixed sources. Only one kind of change clearly raised citation: the reader's own words near the top of the page, by about 0.03 on a 0-to-1 citation score. The words mattered, not the heading form; question headings, lists, cited sources and leading with a direct answer showed no effect larger than 0.03. On 180 queries the first test never used, the gain held (+0.039); a keyword's two words kept part of it, and a question a local model wrote from that keyword did not clearly help. We present Citation Equity, an open-source writing tool rebuilt around this result. It starts from the questions readers ask, shows which of their words a draft lacks near the top, flags settings that keep a page out of answers, and labels the other advice with what our tests found. It runs in the browser and offline.

CCS CONCEPTS

• **Information systems** → **Web searching and information discovery**; • **Human-centered computing** → *Interactive systems and tools.*

KEYWORDS

generative engine optimization, answer engines, citation, writing tools, controlled tests, null results

1 INTRODUCTION

Answer engines, systems that answer a question in prose and cite web sources, now decide which pages a reader sees cited. In a field experiment, a search page with only AI Mode cut click-through

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by 18.8 percentage points [12]. Writers respond with generative engine optimization (GEO): advice and products for getting a page cited.

Most of that advice is untested. Its best-known figure, gains of up to 40%, comes from one setup with five sources in one model's prompt [2]; a survey of 45 studies rejects it as a general claim [8], and adding statistics lowered citation rank in 19 of 24 C-SEO Bench settings [9]. Of 24 commercial products whose public pages we read, most sell brand tracking and seven also rewrite the text,[1] yet writers feel less ownership of text an AI writes for them [3, 4].

We built that advice into a writing tool with eleven checks, then tested each check on its own. An open model made one check's edit to real web pages, a same-length paraphrase of the same passage served as the placebo, and open models answered from fixed sources (80 GEO-bench queries, with 60 from C-SEO Bench as a replication set). Every test was planned in advance. Only the reader's own words near the top of the page clearly raised citation (Figure 1a), and a follow-up showed that the words carry the gain, not the heading. A last test asked what a writer can know: the gain held on 180 queries the first test never used, a keyword's two words kept part of it, and a question a local model wrote from the keyword did not clearly help (Figure 1b).

We present *Citation Equity*, an open-source writing tool rebuilt around these results (Figure 2).[2] It starts from the questions readers ask, counts which ones a draft covers, names the missing words without inserting them, flags settings that keep a page out of answers, and labels the rest of the advice with what our tests found. The name says who it is for: it is free, open and offline, so the evidence is not reserved for brands that pay for tracking tools. Our contributions are (1) a controlled test of eleven common GEO checks, with placebos, a replication set, editor and judge swaps, and tests on new queries and on questions a writer can guess; (2) the finding that vocabulary, not formatting, moved citation; and (3) an open, offline writing tool rebuilt from that finding, with a live demo, code and data.

2 WHAT WE TESTED

The test. For a query and a check, an open model edits only the spans of the target page that the check flags, and validators reject an edit that adds facts or misses the check; the placebo rewords the same spans to the same length. Two open judges answer the query from 4 sources, 8 times each with the order shuffled, and the page's *citation score* is $(5 - r)/4$ when it is the r-th source cited, else 0. A result needs 30 or more queries, tests are Holm-corrected within their family, and "no effect larger than 0.03" means the 90% interval lies inside ±0.03 (plans, code and data: footnote 2).

Eleven checks. Of single changes, only the two that add the reader's own words near the top clearly raised the score (Figure 1a): the reader's question as a section under the title (+0.027, 95% CI [+0.010, +0.043], Holm $p = 0.006$; replication +0.036), and the question's missing words worked into the title and opening (+0.035 against its placebo). The words carry the gain, not the form: the question as a heading did no better than the same text set as plain (+0.003; replication −0.001), while the plain text beat the untouched page (+0.023; +0.037). With another editor and other judges, the reader's-words gain kept its sign (+0.029, $n = 29$). A follow-up added 180 queries for the checks with too few pages: over 260, question headings, cited sources and a direct answer first showed no effect larger than 0.03, as did lists over 80; attributed quotes were inconclusive, and tables, statistics and comparisons still had too few pages.

When the writer has to guess. Every edit above used the reader's exact question, which a writer rarely sees. A last test reran the question section on those 180 new queries, which the first test never used, then built it from two guesses the tool offers: a keyword, here the question's two rarest key words, and a question a local model wrote from it; answers were still judged against the true question. The exact question helped on the new queries (+0.039, 95% CI [+0.023, +0.056], $n = 115$; +0.035 pooled with the first test's queries, $n = 169$), and the reader's missing words worked into the top held there too (+0.036 against their placebo, $n = 74$, shown without a verdict). The keyword helped against the untouched page (+0.016) and against a section without the reader's words (+0.018; Holm $p = 0.006$ for both), but the exact question beat it by +0.019 [+0.008, +0.031], so by the rule we set before the run the keyword loses the gain; on the new queries it kept about two thirds of it. Part of that edge is the exact query string, which the judge's prompt also shows: the exact question beat a section with all its key words but not the string by +0.014. A question a local model wrote from the keyword showed no effect larger than 0.03 (+0.008), and neither did the keyword's missing words worked into the top (+0.004 against their placebo).

> Every gain here tracks the reader's words: once changes in length and in the count of query words are regressed out, what is left includes zero. That is the point for writers: the formatting they are sold did not move citation, and the reader's vocabulary did.

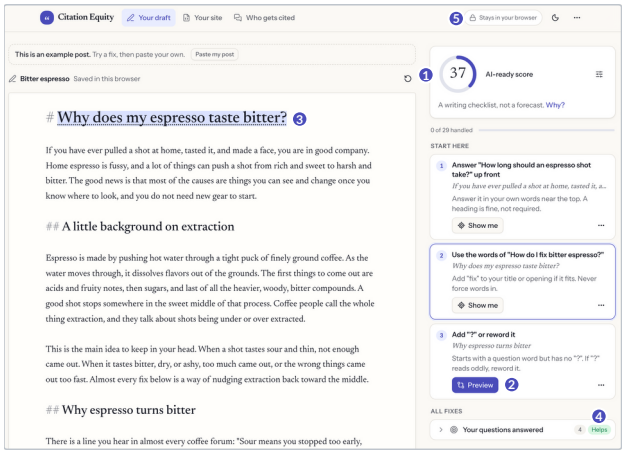
3 CITATION EQUITY

From a score to questions. The first version averaged the eleven checks into a 0-to-100 checklist score. On two sample drafts, following every suggestion raised it by 26 to 29 points, while mean citation share on four open models moved by only 1 to 3 points, and Section 2 found clear gains for only two checks. The tool now shows no total anywhere and starts from the reader's questions.

The flow. Four screens follow the writer's work. *Your questions* asks "What will your readers ask?": the writer types questions, pastes keywords or imports a Search Console or Bing Webmaster export, or loads an example with one click. *Your draft* leads with "Questions covered: N of M" (Figure 2). A question is covered when its key words appear in the title, first heading or first paragraph, and a section opens with a plain sentence that uses most of them; the card names the missing words and never inserts them. Below sit one ready edit and the other checks as collapsed writing notes. *Can engines reach it?* checks pasted HTML and robots.txt for three blockers (a blocked AI crawler, a noindex rule, text that

[1] For example https://www.evertune.ai/pricing and https://writesonic.com/pricing, opened 24 October 2026.

[2] Live demo: https://shehral.github.io/citation-equity/. Code (MIT): https://github.com/shehral/citation-equity. Data: https://huggingface.co/datasets/Shehral/citation-equity-experiments. Research page (plans, results): https://shehral.github.io/citation-equity/research/. Video: https://youtu.be/kbU9t3ewVUk.



1. AI-ready score: "A writing checklist, not a forecast."
2. The ready edit: Preview first, Apply only inside the preview
3. The selected fix's passage, marked in the draft
4. A check's tag from our tests: "Helps"
5. Privacy chip: "Stays in your browser"

Figure 2: The draft view starts from the reader's questions: it counts which ones the draft covers and names the missing words, and shows no score. Espresso example after two typed edits.

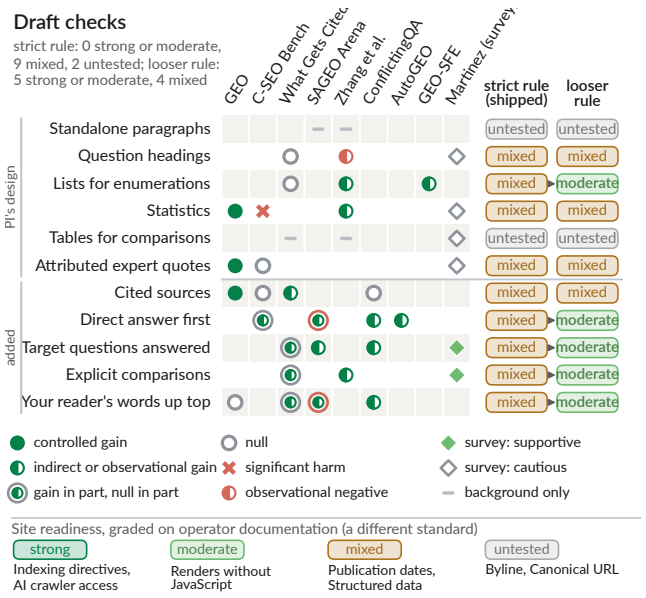


Figure 3: Every draft check has a null or negative result somewhere, but "mixed" spans positive in every model to harmful, and a looser rule, defined after the fact, moves five to moderate. Columns: [2], [9], [10], [6], [15], [11], [13], [14], [8]. One rater.

needs JavaScript), which rest on the engines' own documentation; the other page items are notes. *Did it change anything?* runs the citation test on the draft before and after, and shows whom live AI answers cite.

Labels, not scores. Each check carries one fixed label for what our tests found. A check that helped reads *Helped in our test*, one whose interval sat inside ±0.03 reads *No effect in our test*, and one that ran without a verdict reads *Unclear*, since *Not tested* would be untrue; *Not tested* is kept for inputs our tests never used, such as questions made from the page itself. The two question checks read Helped for the reader's own questions; question headings, lists, cited sources and answer first read No effect; the other five read Unclear. For a question made from a keyword the answer line reads *May help less* and the words line *No effect in our test*, and for a local model's question the answer line reads *No effect in our test*, label words we chose after seeing the results.

Evidence and the rule for nulls. Published studies are graded apart from our labels, on the Research page, under a strict rule: a null or negative result in any study, even in one subgroup, makes a check *mixed*. Over the nine sources the tool cites, each opened and read, no draft check is strong or moderate: nine are mixed and two untested (Figure 3). A looser rule, defined after the fact, that counts only a study's own no-effect conclusion or a significant harm makes five mixed checks moderate. **[OWNER:** Second rater (Ishan Gupta): agreement pending.]

Suggest, never rewrite. Comments anchored to spans of text gave writers more agency than a chat interface in a lab study [7], so every suggestion is anchored to an exact range of the draft and follows the author's edits. An edit is previewed and applied only if its text is unchanged; every action has Undo, and there is no "apply all". An optional local model can propose wording for an answer, applied only on the author's click.

Architecture and the citation test. A TypeScript monorepo holds an *engine* of pure check functions (browser and CLI), a *bench* with the citation test, and a *workspace*, a Next.js static export on GitHub Pages with no server; a browser test asserts that no request leaves the origin on load. For each question the bench retrieves the top 8 of the draft and its competitor pages with BM25, and a local model (`llama3.2:3b` through Ollama) answers with numbered citations under GEO's prompt [2], 8 times with shuffled source orders [8, Sec. 11.2]. Reports keep apart retrieval, valid answers, citation and word share beside the chance level, and each says *"It does not predict live AI engines."* With a sample's 4 questions the smallest possible p is 0.125, so a laptop run shows a direction at most.

4 DEMONSTRATION AT THE VENUE

One laptop serves the static build and runs Ollama (`llama3.2:3b`), so the core demonstration needs no venue network. A second screen shows the hosted demo, and a poster carries Figures 1 and 2.

Station 1, the 90-second loop. The visitor clicks *Try it with an example*: four reader questions, a draft that covers none of them, and one blocker on the example page. Two typed edits, a few words in the title and one answering sentence, take the count to 2 of 4; the visitor then applies the ready edit through Preview, reads its label, and opens the recorded before-and-after test beside its chance line.

A live run (64 answers, 15 minutes) starts each session and fills in while visitors watch. *Station 2, your readers' questions (offline).* Visitors type their questions or keywords and paste their own paragraph; they see which questions it covers and which words are missing, and the text never leaves the laptop. *Station 3, can engines reach your site? (online).* The command-line `ce site` checks the visitor's URL for blockers over a phone hotspot; offline, pasted HTML and robots.txt run the same checks in the browser, except those that need the live site.

5 RELATED WORK

The bench reuses GEO's prompt and word share [2], and our tests follow the counterfactual style of C-SEO Bench and What Gets Cited, which edit one page and hold the rest fixed [9, 10]. Our vocabulary finding fits earlier ones: edits that made a page more relevant to the question made models side with it more often while stylistic edits did not [11], and a page holding the query's terms won the first citation in four of six models [10]. Among open tools, aiseo-audit keeps an evidence tier for every factor and demotes lists and tables to unscored diagnostics [1], and Citation Equity adapts code from it; geo-score keeps measured citation apart from its readiness score [5]. What is new here is a writing tool rebuilt from its own controlled tests, including tests of the questions a writer can actually know.

6 LIMITATIONS AND FUTURE WORK

Every measurement uses open models over fixed sources, not live engines, which differ and drift [8]. The effects are small, about 0.03 on a 0-to-1 score, or about one rank position over eight answers. The keyword in our last test is two words of the real query, so it carries part of the query by design; real keywords may share less. Our first test's queries are those on which the question section first passed, so estimates that include them may run high; the 180 new queries do not have this bias. An earlier sensitivity test of the bench missed its target: a one-sentence answer edit gained +0.007 to +0.034 against a +0.10 bar the harness could detect. The tests cover 320 English queries, nearly all edited by one model family, and the harness prompt has not been byte-checked against the bench's. Next are a comparison with live engines and a study with writers.

7 ETHICS AND USE OF DATA

No human participants took part and no user data was collected. Sample drafts and competitor pages are invented; C-SEO Bench is downloaded at run time, never redistributed; building competitor pages from URLs respects robots.txt. On dual use, the tool writes no page text on its own, asks for real and checkable material, and states the FTC's endorsement rules; the bench could test a manipulative edit, but the tool offers none. The tests use the public pages GEO-bench lists [2]; their text, and everything built from it, is not redistributed, only URLs, hashes and results. A future user study will go through IRB review.

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Use of generative AI. We used Claude (Anthropic) as an assistant for software development, experiment and analysis code, and writing and editing. The authors designed the study, checked the results and take full responsibility for the paper.

REFERENCES

[1] AE Studio. 2026. *aiseo-audit.* Retrieved 2026-10-24 from https://github.com/agencyenterprise/aiseo-audit Commit 6c888aa. MIT License.

[2] Pranjal Aggarwal, Vishvak Murahari, Tanmay Rajpurohit, Ashwin Kalyan, Karthik Narasimhan, and Ameet Deshpande. 2024. GEO: Generative Engine Optimization. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining* (Barcelona, Spain) *(KDD '24)*. ACM, New York, NY, USA, 5–16. https://doi.org/10.1145/3637528.3671900 arXiv:2311.09735 [cs.LG]

[3] Paramveer S. Dhillon, Somayeh Molaei, Jiaqi Li, Maximilian Golub, Shaochun Zheng, and Lionel Peter Robert. 2024. Shaping Human-AI Collaboration: Varied Scaffolding Levels in Co-writing with Language Models. In *Proceedings of the CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) *(CHI '24)*. ACM, New York, NY, USA, 1–18. https://doi.org/10.1145/3613904.3642134

[4] Fiona Draxler, Anna Werner, Florian Lehmann, Matthias Hoppe, Albrecht Schmidt, Daniel Buschek, and Robin Welsch. 2024. The AI Ghostwriter Effect: When Users do not Perceive Ownership of AI-Generated Text but Self-Declare as Authors. *ACM Transactions on Computer-Human Interaction* 31, 2 (2024), 1–40. https://doi.org/10.1145/3637875

[5] Jianrun Tech. 2026. *geo-score.* Retrieved 2026-10-24 from https://github.com/jianruntech/geo-score MIT License.

[6] Sunghwan Kim, Wooseok Jeong, Serin Kim, Sangam Lee, and Dongha Lee. 2026. SAGEO Arena: A Realistic Environment for Evaluating Search-Augmented Generative Engine Optimization. In *Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2* (Jeju Island, Republic of Korea) *(KDD '26)*. ACM, New York, NY, USA, 2342–2353. https://doi.org/10.1145/3770855.3818146 arXiv:2602.12187 [cs.IR]

[7] Martin Lou, Jackie Crowley, Samuel Dodson, and Dongwook Yoon. 2025. AnchoredAI: Contextual Anchoring of AI Comments Improves Writer Agency and Ownership. arXiv preprint. arXiv:2509.16128 [cs.HC]

[8] Olivier Martinez. 2026. Optimizing Visibility in Generative Engines: A Critical Survey of Generative Engine Optimization (2023–2026). arXiv preprint. arXiv:2607.14035 [cs.IR]

[9] Haritz Puerto, Martin Gubri, Tommaso Green, Seong Joon Oh, and Sangdoo Yun. 2025. C-SEO Bench: Does Conversational SEO Work?. In *Advances in Neural Information Processing Systems (NeurIPS 2025), Datasets and Benchmarks Track*. OpenReview.net. arXiv:2506.11097 [cs.CL] https://openreview.net/forum?id=oTeixD3oZO

[10] Rahul Vishwakarma, Shushant Kumar, and Ratnesh Jamidar. 2026. What Gets Cited: Competitive GEO in AI Answer Engines. In *Proceedings of the 49th International ACM SIGIR Conference on Research and Development in Information Retrieval* (Melbourne, VIC, Australia) *(SIGIR '26)*. ACM, New York, NY, USA, 4950–4954. https://doi.org/10.1145/3805712.3808445 arXiv:2605.25517 [cs.AI]

[11] Alexander Wan, Eric Wallace, and Dan Klein. 2024. What Evidence Do Language Models Find Convincing?. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Lun-Wei Ku, Andre Martins, and Vivek Srikumar (Eds.). Association for Computational Linguistics, Bangkok, Thailand, 7468–7484. https://doi.org/10.18653/v1/2024.acl-long.403

[12] Stephanie T. Wang, Jeffrey Gleason, Yakov Bart, Christo Wilson, and Danaé Metaxa. 2026. AI in Search Reduces Publisher Referrals Without Improving User Experience: Experimental Evidence. arXiv preprint. arXiv:2608.18352 [cs.IR]

[13] Yujiang Wu, Shanshan Zhong, Yubin Kim, and Chenyan Xiong. 2026. What Generative Search Engines Like and How to Optimize Web Content Cooperatively. In *The Fourteenth International Conference on Learning Representations (ICLR 2026)*. OpenReview.net. arXiv:2510.11438 [cs.IR] https://openreview.net/forum?id=K8EinVWtUB

[14] Junwei Yu, Mufeng Yang, Yepeng Ding, and Hiroyuki Sato. 2026. Structural Feature Engineering for Generative Engine Optimization: How Content Structure Shapes Citation Behavior. arXiv preprint. arXiv:2603.29979 [cs.CL]

[15] Kai Zhang, Xinyue He, and Jingang Yao. 2026. From Citation Selection to Citation Absorption: A Measurement Framework for Generative Engine Optimization Across AI Search Platforms. arXiv preprint. arXiv:2604.25707 [cs.IR]